CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the series $11 + 17 + 23 + \cdots + 113$ is: 1

(a) 1003 **(b)** 1235 **(c)** 2232 **(d)** 1116

2. How many terms of the AP $18, 15, 12, 9, \ldots$ give sum 0? 1

(a) 12 **(b)** 13 **(c)** 26 **(d)** 14

3. If the 3rd term of an AP is 8 and the 8th term is 3, then the 11th term is: 1

(a) 0 **(b)** 24 **(c)** 11 **(d)** 1

4. The sum of the first 15 terms of an AP is 465, and its first term is 10. The 15th term is: 1

(a) 52 **(b)** 62 **(c)** 72 **(d)** 21

5. In an AP, $S_{12} = 564$ and $S_{11} = 473$. Then $a_{12} =$ 1

(a) 83 **(b)** 99 **(c)** 91 **(d)** 1037

6. The sum of the first n terms of an AP is $S_n = 4n^2 + 5n$. Its common difference is: 1

(a) 4 **(b)** 9 **(c)** 5 **(d)** 8

7. The sum of the series $(-8) + (-5) + (-2) + \cdots + 79$ is: 1

(a) 2130 **(b)** 1065 **(c)** 986 **(d)** 1147

8. How many terms of the AP $30, 25, 20, 15, \ldots$ give sum 0? 1

(a) 12 **(b)** 13 **(c)** 26 **(d)** 14

9. **Assertion (A):** The sequence $1, 4, 9, 16, \ldots$ is an AP. 1

Reason (R): In an AP, the difference between consecutive terms is constant.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The sum of the first 10 natural numbers is 55. 1
Reason (R): The nth term of an AP is $a_n = a + (n-1)d$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $S_n = 2n^2 + 5n$, find the first term and the common difference. 2

12. In an AP, $a_3 + a_7 = 36$ and $a_2 + a_9 = 40$. Find a and d. 2

13. The 6th term of an AP is 11 and the 11th term is 6. Find the 17th term. 2

14. If $S_n = 4n^2 + 1n$, find the first term and the common difference. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, $S_7 = 98$ and $S_{12} = 228$. Find the AP. 3

16. Find the sum of the first 14 terms of the AP whose 4th term is 21 and 9th term is 36. 3

17. Find the sum of all integers between 5 and 150 that leave remainder 1 when divided by 6. 3

OR

In an AP, $S_4 = 42$ and $S_8 = 164$. Find the AP. 3

18. Find the sum of the first 18 terms of the AP whose 3rd term is 18 and 6th term is 39. 3

OR

Find the sum of all integers between 30 and 400 that leave remainder 5 when divided by 11. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** The school auditorium has 15 rows of seats. The first row has 20 seats and each next row has 2 more seats than the row before it. 5
(i) Write the number of seats in the first three rows.
(ii) How many seats are there in the last (15th) row?
(iii) Find the total number of seats in the auditorium.
(iv) Find the number of seats in the 8th row and the total collection if every seat is sold at ₹ 150.

20. A ladder has 19 rungs whose lengths decrease uniformly from 120 cm at the bottom to 48 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

OR

A sum of ₹1150 is to be given as 10 prizes, each prize ₹30 less than the preceding one. Find the value of the first and last prize. 5

END OF QUESTION PAPER — Total: 40 Marks